

Week 2 · Frame It as Cases

Muhrram Ali · AI Fluency

Voice Card

direct, plain, honest, specific, no buzzwords

(Standing instruction for every draft: use short sentences, my real words, and never add “passionate”, “results-driven”, or generic AI filler.)

Case 1 — Taking Models into Working Pipelines

(AI Intern · Cosmic.AI 365)

The problem

The team had working experiments and notebooks, but turning them into something other people could actually use was slow and messy. Models stayed stuck in the experimental stage.

What I did

I took models from pure experimentation into production-facing pipelines. I connected them to dashboards, automated parts of the workflow, and handled text-to-speech pieces. I also cleaned up the user experience so the tools were easier to run day-to-day.

What came of it

The models moved from notebooks into systems people could actually use. The internship confirmed I can close the gap between “it works on my machine” and “it runs for others.” That gap is still the part of AI work I care about most.

Case 2 — YOLO Object Detection Pipeline

The problem

I wanted to understand not just how to run an object detector, but what happens when you have to choose between accuracy, speed, and what you can actually deploy.

What I did

I built a YOLO-based detection pipeline end to end. I worked through image preprocessing, inference, and confidence scoring. Then I deliberately compared the trade-offs: higher accuracy versus faster inference versus the constraints of a real deployment target.

What came of it

I stopped treating detection accuracy as the only number that matters. The useful skill turned out to be knowing which trade-off is acceptable for the situation. That mindset now shows up in how I approach any model I put into a pipeline.

Case 3 — Task API (CRUD)

(Built during this internship · Week 2)

The problem

I needed to prove I can build a complete, correct backend from scratch — the kind of small system every real product depends on — without a database and with proper status codes and validation.

What I did

I built a full in-memory To-Do API with FastAPI. I implemented all four CRUD operations, correct HTTP status codes (201, 204, 400, 404), input validation, and Swagger documentation. I also added simple filtering, a stats endpoint, and a reset route so the API stays usable for demos.

What came of it

A working public API that anyone can clone and run in a few minutes. More importantly, I now have a concrete reference for every backend I meet: the same four operations, the same status-code discipline, and the same habit of never trusting the client.

Repo: <https://github.com/muhrramali/todo-api>

Bio

I take machine-learning models from experimentation into working pipelines that respect real systems and deployment constraints. I'm a final-year Computer Systems Engineering student looking for research or internship opportunities where that skill is useful.

Contact / CTA

Email me if you're a research lead or PI who needs an undergraduate already able to ship end-to-end AI systems: muhrram008@gmail.com

Before / After Example

Generic AI version (before)

"Leveraged cutting-edge deep learning techniques to deliver impactful computer vision solutions and drive measurable results in a fast-paced environment."

My edited version (after)

"I built a YOLO detection pipeline and deliberately measured the trade-offs between accuracy, speed, and what I could actually deploy. That comparison changed how I think about putting any model into production."

This document uses only work I actually did. Every case follows the three beats and points toward the same audience and action from Week 1.